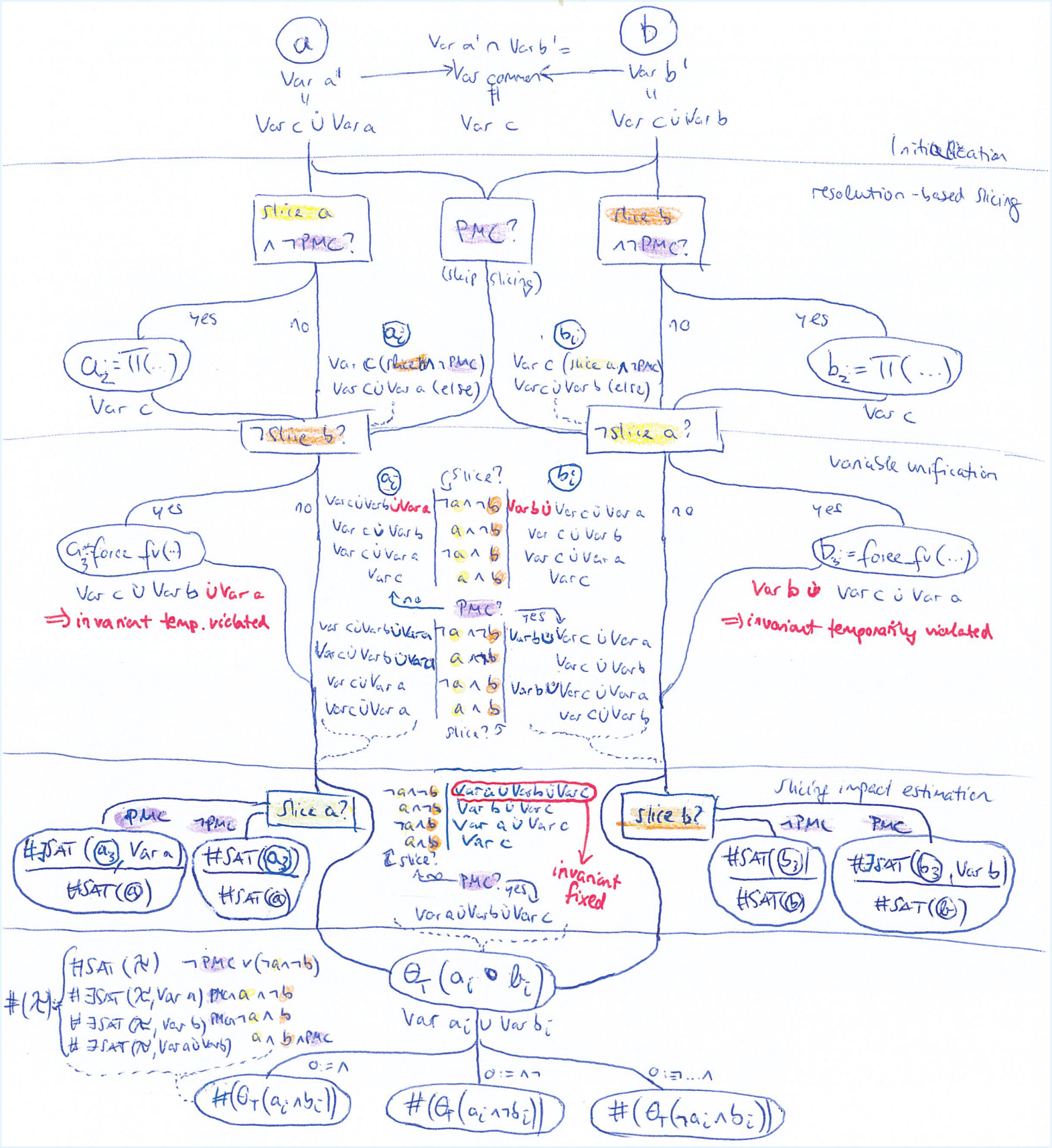
a
b
Var a' ∩ Var b' =
Var a'
Var common
Var b'
Var c ∪ Var a
Var c
Var c ∪ Var b
Initialization
resolution-based slicing
slice a ∧ ¬PMC?
PMC?
slice b ∧ ¬PMC?
(skip slicing)
yes
no
a_i := Π(...)
Var c
a_i
Var c (slice a ∧ ¬PMC)
Var c ∪ Var a (else)
b_i
Var c (slice a ∧ ¬PMC)
Var c ∪ Var b (else)
b_i := Π(...)
Var c
¬slice b?
¬slice a?
variable unification
a_i
slice?
b_i
Var c ∪ Var b ∪ Var a
Var c ∪ Var b
Var c ∪ Var a
Var c
¬a ∧ ¬b
a ∧ ¬b
¬a ∧ b
a ∧ b
Var b ∪ Var c ∪ Var a
Var c ∪ Var b
Var c ∪ Var a
Var c
a_i := force_fv(...)
Var c ∪ Var b ∪ Var a
⇒ invariant temp. violated
b_i := force_fv(...)
Var b ∪ Var c ∪ Var a
⇒ invariant temporarily violated
PMC?
Var c ∪ Var b ∪ Var a
Var c ∪ Var b ∪ Var a
Var c ∪ Var a
Var c ∪ Var a
¬a ∧ ¬b
a ∧ ¬b
¬a ∧ b
a ∧ b
Var b ∪ Var c ∪ Var a
Var c ∪ Var b
Var b ∪ Var c ∪ Var a
Var c ∪ Var b
slice?
slice impact estimation
slice a?
PMC
¬PMC
#∃SAT(a_3, Var a) / #SAT(a)
#SAT(a_3) / #SAT(a)
¬a ∧ ¬b
a ∧ ¬b
¬a ∧ b
a ∧ b
Var a ∪ Var b ∪ Var c
Var b ∪ Var c
Var a ∪ Var c
Var c
invariant fixed
slice?
PMC?
Var a ∪ Var b ∪ Var c
slice b?
¬PMC
PMC
#SAT(b_3) / #SAT(b)
#∃SAT(b_3, Var b) / #SAT(b)
#(φ) =
#SAT(φ) ¬PMC ∨ (¬a ∧ ¬b)
#∃SAT(φ, Var a) PMC ∧ a ∧ ¬b
#∃SAT(φ, Var b) PMC ∧ ¬a ∧ b
#∃SAT(φ, Var a ∪ Var b) a ∧ b ∧ PMC
θ_T(a_i ∘ b_i)
Var a_i ∪ Var b_i
∘ := ∧
∘ := ∧¬
∘ := ∃...∧
#(θ_T(a_i ∧ b_i))
#(θ_T(a_i ∧ ¬b_i))
#(θ_T(¬a_i ∧ b_i))